



JKBOSE 2026

CLASS 11TH

SEQUENCES AND SERIES

MATHS

LECTURE-02

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TOPICS TO BE COVERED

1

Geometric Progression

2

Related Problems

Geometric Progression (G.P.)

$$\begin{array}{ccccccccc}
 a_1 & & a_2 & & a_3 & & a_4 & & a_5 \\
 2 & , & 4 & , & 8 & , & 16 & , & 32 , \dots
 \end{array}$$

$\xrightarrow{\times 2} \quad \xrightarrow{\times 2} \quad \xrightarrow{\times 2} \quad \xrightarrow{\times 2}$ (G.P.)

* A sequence non-zero terms whose each succeeding term can be obtained from the preceding term by multiplying with a constant is known as a Geometric progression (G.P.)

* Constant is generally denoted by δ and is known as **Common ratio**.

$$* \quad 1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots$$

$\xrightarrow{\times \frac{1}{2}} \quad \xrightarrow{\times \frac{1}{2}} \quad \xrightarrow{\times \frac{1}{2}} \quad \xrightarrow{\times \frac{1}{2}}$

$\Rightarrow$ It is a G.P. with $\delta = \frac{1}{2}$.

* In a G.P., the ratio of successive terms is constant.

i.e. $\frac{a_n}{a_{n-1}} = \delta (\text{Constant})$

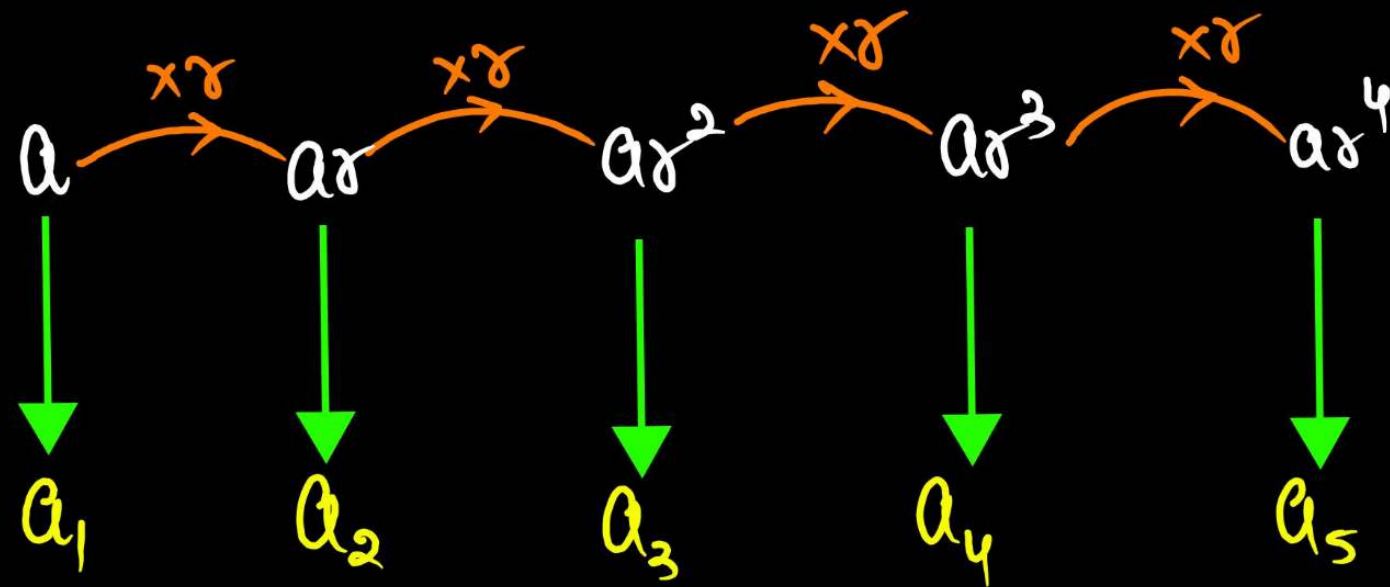
$$\begin{array}{ccccccccc}
 a_1 & & a_2 & & a_3 & & a_4 & & a_5 \\
 3 & , & 9 & , & 27 & , & 81 & , & 243 , \dots
 \end{array}$$

$\xrightarrow{\times 3} \quad \xrightarrow{\times 3} \quad \xrightarrow{\times 3} \quad \xrightarrow{\times 3}$ (Geometric Progression)

$$\delta = 3 = \frac{a_2}{a_1} = \frac{a_3}{a_2} = \frac{a_4}{a_3} = \frac{a_5}{a_4}$$



General Term of a G.P.:



Based on the above observation, we can conclude that;

$$a_n = ar^{n-1} \text{ (n}^{\text{th}} \text{ term of the G.P.)}$$

Sum of n-Terms of a G.P.:

Let S_n be the sum of n-Terms of a G.P. Then;

$$S_n = a + ar + ar^2 + ar^3 + \dots + ar^{n-2} + ar^{n-1} \quad (1)$$

Suppose that $r = 1$, then

$$S_n = a + a + a + a + \dots + a + a$$

$$\Rightarrow S_n = na$$

Also, suppose that $r \neq 1$, then multiplying by r on b/s of (1), we get;

$$rS_n = ar + ar^2 + ar^3 + ar^4 + \dots + ar^{n-1} + ar^n \quad (2)$$

Subtracting ① & ②, we get;

$$S_n - rS_n = (a + ar + ar^2 + ar^3 + \dots + ar^{n-1}) - (ar + ar^2 + ar^3 + \dots + ar^{n-1} + ar^n)$$

$$\Rightarrow S_n(1-r) = a - ar^n$$

$$\Rightarrow S_n(1-r) = a(1-r^n)$$

$$\Rightarrow S_n = \frac{a(1-r^n)}{1-r} = \frac{a(r^n-1)}{r-1}$$

So,

$$S_n = \begin{cases} \frac{a(r^n-1)}{r-1}, & r \neq 1 \\ na & ; r = 1 \end{cases}$$

EXAMPLE

Find the 10th and nth terms of the G.P. 5, 25, 125,

(Sol) Given G.P. is:

$$5, 25, 125, \dots$$

$$\text{i.e. } 5, 5^2, 5^3, \dots$$

$$\text{Here } a = 5, r = 5.$$

$$\text{So } a_n = ar^{n-1} = 5 \times 5^{n-1}$$

$$\Rightarrow a_n = 5^n$$

$$\text{Also, } a_{10} = 5^{10}.$$

EXAMPLE

$$a^x = a^y \Rightarrow x = y \\ a \neq 0, 1, -1$$



How many terms of the G.P. $3, \frac{3}{2}, \frac{3}{4}, \dots$ are needed to give the sum $\frac{3069}{512}$?

(Sol) Given G.P. is:

$$3, \frac{3}{2}, \frac{3}{4}, \dots$$

$$\text{Here } a = 3, r = \frac{1}{2}.$$

$$\text{let } S_n = \frac{3069}{512}.$$

$$\Rightarrow \frac{a(1-r^n)}{1-r} = \frac{3069}{512}$$

$$\Rightarrow \frac{3\left(1 - \frac{1}{2^n}\right)}{\left(1 - \frac{1}{2}\right)} = \frac{3069}{512}$$

$$\Rightarrow 2\left(1 - \frac{1}{2^n}\right) = \frac{1023}{512}$$

$$\Rightarrow 1 - \frac{1}{2^n} = \frac{1023}{1024}$$

$$\Rightarrow \frac{1}{2^n} = 1 - \frac{1023}{1024}$$

$$\Rightarrow \frac{1}{2^n} = \frac{1024 - 1023}{1024}$$

$$\Rightarrow \frac{1}{2^n} = \frac{1}{2^{10}}$$

$$\Rightarrow 2^n = 2^{10}$$

$$\Rightarrow \boxed{n = 10}$$

EXAMPLE

The sum of first three terms of a G.P. is $\frac{13}{12}$ and their product is -1 . Find the common ratio and the terms.

(Sol) let the required three terms be $\frac{a}{r}, a, ar$. Then According to the

Question;

$$\frac{a}{r} + a + ar = \frac{13}{12}$$

$$\Rightarrow a\left(\frac{1}{r} + r + 1\right) = \frac{13}{12} \quad \text{--- (1)}$$

$$\text{Also } \frac{a}{r} \times a \times ar = -1.$$

$$\Rightarrow a^3 = -1.$$

$$\Rightarrow a = -1 \quad \text{--- (2)}$$

Using (2) in (1), we get:

$$-\left(\frac{1}{r} + r + 1\right) = \frac{13}{12}$$

$$\Rightarrow \frac{r^2 + r + 1}{r} = -\frac{13}{12}$$

$$\Rightarrow 12r^2 + 12r + 12 = -13r$$

$$\Rightarrow 12r^2 + 25r + 12 = 0$$

$$\Rightarrow 12r^2 + 16r + 9r + 12 = 0$$

$$\Rightarrow 4r(3r + 4) + 3(3r + 4) = 0$$

$$\Rightarrow (4r + 3)(3r + 4) = 0.$$

$$\Rightarrow r = -\frac{3}{4}, -\frac{4}{3}.$$

∴ the required G.P.'s are, $\frac{4}{3}, -1, \frac{3}{4}$ or $\frac{3}{4}, -1, \frac{4}{3}$

Hence the result.

EXAMPLE

Find the sum of the sequence $7, 77, 777, 7777, \dots$ to n terms.

(Sol) let $S_n = 7 + 77 + 777 + \dots$ upto n terms

$$\Rightarrow S_n = 7(1 + 11 + 111 + \dots \text{upto } n \text{ terms})$$

$$\Rightarrow S_n = \frac{7}{9} \times 9(1 + 11 + 111 + \dots \text{upto } n \text{ terms})$$

$$\Rightarrow S_n = \frac{7}{9}(9 + 99 + 999 + 9999 + \dots \text{upto } n \text{ terms})$$

$$\Rightarrow S_n = \frac{7}{9} \left\{ (10-1) + (10^2-1) + (10^3-1) + (10^4-1) + \dots + (10^n-1) \right\}$$

$$\Rightarrow S_n = \frac{7}{9} \left\{ (10 + 10^2 + 10^3 + \dots + 10^n) - (1 + 1 + 1 + 1 + \dots + 1) \right\}$$

$$\Rightarrow S_n = \frac{7}{9} \left\{ \frac{10(10^n-1)}{10-1} - n \right\}$$

$$\Rightarrow S_n = \frac{7}{9} \left\{ \frac{10^{n+1}-10}{9} - n \right\}$$

$$\Rightarrow S_n = \frac{7}{9} \left(\frac{10^{n+1}-10-9n}{9} \right)$$

$$\Rightarrow S_n = \frac{7}{81} (10^{n+1} - 9n - 10)$$

QUESTION (NCERT EXAMPLAR)

The third term of G.P. is 4. The product of its first 5 terms is

1

$4^3 \times$

(Sol) $ar^2 = 4$

2

$4^4 \times$

So, $a \times ar \times ar^2 \times ar^3 \times ar^4$

$= a^5 r^{10}$

$= (ar^2)^5$

$= 4^5$

3

$4^5 \checkmark$

4

None of these $\times$

QUESTION

If $3, 3^2, 3^3$ are in G.P., then $\log 3, \log 9$ and $\log 27$ are in

1

AP ✓

$$\begin{array}{ccc} \log 3, & 2\log 3, & 3\log 3 \\ a_1 & a_2 & a_3 \end{array}$$

2

GP ✗

$$a_2 - a_1 = \log 3$$

$$a_3 - a_2 = \log 3$$

3

HP ✗

4

None of these ✗

QUESTION (JEE MAINS 2013)

The sum of first 20 terms of the sequence $0.7, 0.77, 0.777, 0.7777, \dots$ is

1 $\frac{7}{81}(179 - 10^{-20})$

2 $\frac{7}{9}(99 - 10^{-20})$

3 $\frac{7}{81}(179 + 10^{-20})$

4 $\frac{7}{9}(99 + 10^{-20})$

Sol) let $S_{20} = 0.7 + 0.77 + 0.777 + \dots$ upto 20 terms

$\Rightarrow S_{20} = \frac{7}{10} + \frac{77}{10^2} + \frac{777}{10^3} + \dots$ upto 20 terms

$\Rightarrow S_{20} = \frac{7}{9} \left(\frac{9}{10} + \frac{99}{10^2} + \frac{999}{10^3} + \dots \right)$ upto 20 terms

$\Rightarrow S_{20} = \frac{7}{9} \left\{ \left(\frac{10-1}{10} \right) + \left(\frac{10^2-1}{10^2} \right) + \left(\frac{10^3-1}{10^3} \right) + \dots + \left(\frac{10^{20}-1}{10^{20}} \right) \right\}$

$\Rightarrow S_{20} = \frac{7}{9} \left\{ \left(1 - \frac{1}{10} \right) + \left(1 - \frac{1}{10^2} \right) + \left(1 - \frac{1}{10^3} \right) + \dots + \left(1 - \frac{1}{10^{20}} \right) \right\}$

$$\Rightarrow S_{20} = \frac{7}{9} \left\{ 20 - \left(\frac{1}{10} + \frac{1}{10^2} + \frac{1}{10^3} + \dots + \frac{1}{10^{20}} \right) \right\}$$

$$\Rightarrow S_{20} = \frac{7}{9} \left\{ 20 - \frac{\frac{1}{10} \left(1 - \frac{1}{10^{20}} \right)}{\left(1 - \frac{1}{10} \right)} \right\} = \frac{7}{9} \left\{ 20 - \frac{\cancel{\frac{1}{10}} \left(1 - \frac{1}{10^{20}} \right)}{\cancel{\frac{9}{10}}} \right\}$$

$$\Rightarrow S_{20} = \frac{7}{9} \left\{ \frac{180 - 1 + 10^{-20}}{9} \right\}$$

$$\Rightarrow S_{20} = \frac{7}{81} (179 + 10^{-20})$$

QUESTION (NCERT EXAMPLAR)

If $x, 2y, 3z$ are in A.P., where the distinct numbers x, y, z are in G.P. then the common ratio of the G.P. is

1 3

2 $\frac{1}{3}$

3 2

4 $\frac{1}{2}$

(Homework Problem)

QUESTION (NCERT EXAMPLAR)

If the 2nd, 5th and 9th terms of a non-constant A.P. are in G.P, then the common ratio of this G.P. is:

1 $\frac{4}{3}$

2 1

3 $\frac{7}{4}$

4 $\frac{8}{5}$

(Homework Problem)

QUESTION (NCERT EXAMPLAR)

The length of three unequal edges of a rectangular solid block are in G.P. The volume of the block is 216 cm^3 and the total surface area is 252 cm^2 . The length of the longest edge is.

1 12 cm

2 6 cm

3 18 cm

4 3 cm

(Homework
problems)

Thank
You